



Enter

Judge



About



Entries



Action tasks

4

Chroma-Scan

ID

7629-CPM

Entry name

Chroma-Scan

Category

Chemical, Physical and Mathematical Sciences

Student 1

Full Name

Protas Mayaka

Gender



Male



Female

Date of Birth

2008-05-29

Kindly use the following format to enter your birth date: dd/mm/yyyy format

Class

Form 4

Student 2:

Full Name 2 (optional)

Graffins Onyangi

Gender 2 (optional)

☒ Male

☐ Female

Date of Birth 2 (optional)

2008-11-13

Class 2 (optional)

Form 4

Supervising Teacher:

Teacher Full Name

BON ONCHARI

Teacher Gender

☒ Male

☐ Female

Student 1 Picture

Select file

The student's picture should be presentable - do not upload a selfie, or a picture with filters, or one with a pose. This picture will be used on YSK platforms.

Student 2 Picture

Select file

The student's picture should be presentable - do not upload a selfie, or a picture with filters, or one with a pose. This picture will be used on YSK platforms

Section 1: Special Award Self Nomination

Kindly nominate your project for a Special Award for it to be considered during the judging process (optional)

Innovative Project on STEM Inclusivity



Section 2: Project Information

The solution

0 / 300 words

In Kenyan secondary schools, quantitative chemical analysis is severely hindered by a "Scientific Equipment Gap". Commercial spectrophotometers routinely cost over KSH 100,000, forcing students to rely on subjective visual titration methods that limit precision during critical chemistry practicals. The Chroma-Scan is highly significant as it democratizes advanced chemical analysis by providing a digital spectrophotometer for under KSH 4,000. Based on the Beer-Lambert Law, it measures light absorbance to calculate molar concentrations. To bypass the non-linear noise and inaccuracy of cheap electronic sensors, the system integrates a Machine Learning Regression layer, providing university-level analytical tools using highly affordable, open-source STEM hardware.

Explain the relevance of the topic, including the problem statement. Give a brief background on who has been studying this area. Explain why is doing this work was important or who cares about it and why? Explain what is the significance of their work within the larger research, technology, science, social context.

Materials and Methods

0 / 300 words

The device was built using affordable microelectronics and standard chemical reagents.

Hardware Assembly: We 3D-printed a light-isolated chamber designed to hold standard 10mm cuvettes. An RGB LED (Common Cathode) was used as a quasi-monochromatic light source, pulsing complementary colors to maximize absorbance. A high-precision Light Dependent Resistor (LDR) captured the transmitted light.

Data Processing: An Arduino Nano processed the analog voltage signals from the LDR and displayed real-time molarity on a 128x32 OLED screen.

AI Neural Calibration: Instead of a standard linear formula, the system was trained on standard serial dilutions of Copper (II) Sulfate and Potassium Permanganate. A Machine Learning Regression layer mathematically analyzed this dataset to learn and "filter out" the non-linear electronic noise profile specific to the LDR.

Describe in detail, the materials/reagents that were used in the experiment. Give a detailed description of the methodology that was used, the exact protocols that were followed, including the materials/reagents that were used.

Results

0 / 300 words

Testing the Chroma-Scan device against standard serially diluted chemical solutions yielded exceptional precision metrics.

Linearity: The AI-calibrated output generated a highly linear calibration curve that strictly verified the Beer-Lambert Law. The coefficient of determination (R^2) for the regression model was recorded at >0.985 , indicating a remarkably strong correlation.

Error Reduction: During stress testing under varying ambient light conditions, the Neural Calibration algorithm successfully filtered out baseline drift, reducing the error margin by 15% compared to raw digital sensors operating without ML processing.

Accuracy: For unknown sample solutions, the device consistently achieved an overall concentration prediction accuracy exceeding 95%.

The results should be an exact description of what was obtained and not what the student envisioned to get. The results should be interpreted in the form of graphs, tables and other forms used to describe results.

Conclusions and Recommendations

0 / 300 words

We concluded that AI-enhanced hardware can successfully democratize advanced chemistry analysis. By mathematically bypassing the limitations of low-cost sensors using intelligent software, Chroma-Scan enables underfunded public schools to conduct high-fidelity quantitative analysis.

Limitations: The LDR struggles with subtle pastels and dark shades due to a low signal-to-noise ratio.

Recommendations: Future iterations should implement an advanced AS7262 6-channel visible light spectral sensor for true spectral data. We also recommend integrating a K-Nearest Neighbors (KNN) color classification algorithm and a micro-SD module for automated long-term laboratory data logging during reaction rate experiments.

What conclusions can the students come up with? What recommendations can they make, for example, should there be further testing? What limitations did they encounter? Should the technology/research be rolled out? Should the Government/industry make specific recommendations?

Does your solution already exist? [Yes/No]

If yes, tell us what is unique/different about your project? If no, tell us your creative solution.

0 / 300 words

Yes, commercial digital spectrophotometers exist, but they cost over KSH 100,000. While there are low-cost "DIY" colorimeters, they are notoriously inaccurate because cheap Light Dependent Resistors (LDRs) do not respond to light linearly, violating the math required for the Beer-Lambert Law. Chroma-Scan's unique, creative solution is "Neural Calibration". We use a Machine Learning Regression algorithm as a "software fix" for a "hardware limitation." The AI is trained on standard chemical solutions to learn the specific non-linear electronic noise pattern of the cheap LDR, mathematically filtering it out to achieve a 95% prediction accuracy.

To help with protecting your intellectual property, we'd like to better understand if your project is new and unique or if there are other solutions that exist and what makes your project unique from already existing solutions.

Section 3: Project Attachments

Project Report Book

Select file

The project report book should be 15-38 pages depending with the depth of your project research.
Preferably 20 - 25 pages.

Project Display

Select file

Your project display should be a summary of your project; only cover the main points and highlights of the project

Video Presentation

Select file

Your video should be a maximum of 4 minutes

Project Log Book (optional)

Select file

This is a record of the number of tries/experiments that you carried out as you worked on your project

- ☒ We confirm that this project is our own original work and is not a reproduction of any other published or presented work from any other source, local or international. The results presented herein were obtained from experiments, research or trials that we have undertaken ourselves and this is the first time that these results are being presented to the Young Scientists Kenya exhibition for judgement.

Section 4: Team T-shirt Sizes

Enter Student 1 T-shirt Size:

L

Enter Student 2 T-shirt size: (optional)

L

Enter Teacher T-shirt size:

XL

Abstract

Results / Conclusions: The Chroma-Scan achieved a linear calibration curve with an

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